

API Specifications

LAMA

Document Version 1.3

API Version V1

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National Stock Exchange of India Ltd
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LAMA APIs:

The base URL for the LAMA (Log Analytics and Monitoring Application) APIs would be as below

UAT: <https://lama.uat.nseindia.com/api/{version}>

E.g.: <https://lama.uat.nseindia.com/api/V1>

Below are the UAT URLs for LAMA API endpoints:

- a. Login: <https://lama.uat.nseindia.com/api/V1/auth/login>
- b. Logout: <https://lama.uat.nseindia.com/api/V1/auth/logout>
- c. Application Metrics: <https://lama.uat.nseindia.com/api/V1/metrics/application>
- d. Hardware Metrics: <https://lama.uat.nseindia.com/api/V1/metrics/hardware>
- e. Network Metric: <https://lama.uat.nseindia.com/api/V1/metrics/network>
- f. Database Metrics: <https://lama.uat.nseindia.com/api/V1/metrics/database>

Note:

- 1) The version number in the UAT URL will be the [API Version No](#) specified at the beginning of the document
- 2) The LAMA APIs are to be executed at an interval of every 5 mins for 24 hours on all trading days.
- 3) For all the APIs the fields in the request body (JSON) are kept fixed i.e. invalid characters and spaces would not be allowed.

Following are the API endpoints:

Sr.no	Method	API EndPoint	Remarks
1	POST	/auth/login	To login: e.g. /auth/login
2	POST	/auth/logout	To logout: e.g. /auth/logout
3	POST	/metrics/application	To publish application metrics: /metrics/application
4	POST	/metrics/hardware	To publish hardware metrics: /metrics/hardware
5	POST	/metrics/database	To publish database metrics: /metrics/database
6	POST	/metrics/network	To publish application network: /metrics/network

Login

This API validates the user credentials. On successful login, a unique token will be provided per request and session will be established with user's system. In subsequent API calls, this token should be passed in HTTP header "Authorization".

For Password:

- The password needs to be sent in a hash string
- The password needs to be sent in encrypted format using AES256. The encryption private key will be generated on user registration
- After 5 invalid attempts with wrong password, the user will not be allowed to generate the token by calling the API.
- The code snippet for password encryption is attached in the below document.



AES

PasswordEncryption.c

For Token:

- Login session validity will be for 24 hours.
- After every 24 hours the token will expire, and the approach is that of single token and not refresh token.

Method	API EndPoint	Header	Content-Type
POST	/auth/login		application/json

Request Object values

Sr. No.	Field Description	Field Name	Data Type	Description	Optional /Mandatory
1	Member ID	memberId	String	Member Code	M
2	Login ID	loginId	String	Login ID	M
3	Password	password	Char(50)	Password should be encrypted with AES256 encryption using the secret key received during API registration	M

Request Object Structure (JSON)

```
{
  "memberId": "06000",
  "loginId": "ApiUser06000",
  "password": "0CxUgThnqIK6gaKU2HnL2A=="
}
```

Password in example is encrypted text

Response JSON values

Sr. No.	Field Description	Field Name	Data Type	Description
1	Timestamp	timestamp	Numeric	Timestamp (in milliseconds since NSE epoch time i.e., 01-jan-1980)
2	Version No	versionNo	String	API Version
3	Member ID	memberId	String	Member Code
4	Login ID	loginId	String	Login ID
5	Response Code	responseCode	Char(3)	Response Code
6	Response Description	reponseDesc	String	Login status description
7	Token	token	String	Unique Token

Response Object Structure (JSON)

```
{
  "timestamp": 1356998400000,
  "versionNo" : "V1",
  "memberId": "09100",
  "loginId": "ApiUser09100",
  "responseCode": "601",
  "responseDesc": "success",
  "token": "eyJhbGciOiJSUzI1NiJ9.eyJzdWUiOiIwOTUwMCImlzcyI6IjFkFwaVVzZXIwOTUwMCImlmV4cCI6MjYzYzYyOCwiaWF0IjoxNjMyMjA0MDI4LCJqdGkiOiJhNWUyYjZiZi1lZTQwLTQ4NzltOGM4ZC0xNjE5ZjA2Nzk2NTEifQ.gGmBfplbtP59X6gK28z4l6lIkFgOBOIGjAX-WHjkY5DPJRxHJWtC-TOcDyPjLGAvnHNlvh-SZRRawkxFSNgYw"
}
```

Logout

This API closes the current session and gets the user logged off.

Method	API EndPoint	Header	Content-Type
POST	/auth/logout	Session Token	application/json

Request JSON values

Sr. No.	Field Description	Field Name	Data Type	Description
1	Member ID	memberId	Char(5)	Member Code
2	Login ID	loginId	Char(16)	Login ID

Request Object Structure (JSON)

```
{
  "memberId": "06000",
  "loginId": "ApiUser06000"
}
```

Response JSON values

Sr. No.	Field Description	Field Name	Data Type	Description
1	Timestamp	timestamp	Numeric	Timestamp (in milliseconds since NSE epoch time i.e., 01-jan-1980)
2	Version No	versionNo	String	API Version
3	Response Code	responseCode	Char(3)	Response Code
4	Response Description	reponseDesc	String	Logout status description

Response Object Structure (JSON)

```
{
  "timestamp": 1356998400000,
  "versionNo": "V1",
  "responseCode": "601",
  "responseDesc": "success"
}
```

Value Table

Field Description	Field Name	Values
Exchange ID	exchangeId	-1=Not Applicable 1= National Stock Exchange 2=Bombay Stock Exchange 3=Metropolitan Stock Exchange 4=Multi-Commodity Exchange 5=National Commodity and Derivatives Exchange
Application ID	applicationId	-1=Not Applicable 1= Client Connectivity 2= Order Management System 3= Risk Management System 4= Exchange Connectivity

Error Table

Field Description	Field Name	Data type	Description
Application ID	applicationId	Numeric (1)	Application ID for the Record
Error Code	errCode	Numeric (3)	Validation Error Code
Error Description	errDesc	String (500)	Description of the error code
Error Key	errKey	String (100)	The incorrect/invalid metric key
Measure	measure	String (3)	The incorrect measure value

Measure Table

Field Description	Field Name	Data Type
Minimum	min	Numeric (3,2)
Maximum	max	Numeric (3,2)
Average	avg	Numeric (3,2)
Median	med	Numeric (3,2)

Log Table

Field Description	Data Type	Expected Values
Time Stamp (in milliseconds since NSE epoch time i.e., 01-jan-1980)	Numeric (13,0)	1356998400000
Server Name	String (50)	Appserver1
Source Name	String (50)	172.3.2.1
Log ID	Long	1245
Log Level (ERROR, WARNING, FATAL)	String (5)	ERROR, WARN, FATAL
Description	String (2000)	Description1

Application Metrics

This API will publish the different application metrics of the member end systems.

Method	API EndPoint	Header	Content-Type
POST	/metrics/application	Session Token	application/json

Request Object values

Field Description	Field Name	Data Type	Description	Expected Values & Validations	Optional/ Mandatory
Payload	payload	List	List of application records	List of payload	M
Exchange ID	exchangeId	Numeric	Different exchange ids	Refer Value table	M
Member ID	memberId	String	Member code	09100	M
Sequence ID	sequenceId	Numeric	Sequence number of the records	1	M
Location ID	locationId	Numeric (6)	Location Identifier number	1	O
Time Stamp (in milliseconds since NSE epoch time)	timestamp	Numeric	Time stamp of the records	1356998400000	M

i.e., 01-jan-1980)					
--------------------	--	--	--	--	--

Payload

Field Description	Field Name	Data Type	Description	Expected Values & Validations	Optional/ Mandatory
Application ID	applicationId	Numeric	Application Identification number	1	M
Metric Data	metricData	List	List of key value pair with all available metrics in key and their corresponding values	<pre>[{"key":"","value":""}]</pre> Available Keys: 1. log 2. throughput 3. latency 4. failureTradeApi 5. failureAuthentication 6. historicalThroughput 7. historicalLatency	M

Metric Data

Field Description	Key Name	Data Type	Frequency	Expected Value
Log Monitoring (log messages for log levels Error, Fatal and Warning)	log	String (2000)	Every 5 min	Timestamp, Server Name, Source Name, Log ID, Log Level, Description CSV value refer the log table
Request/Second - throughput of E2E system	throughput	Object	Every 5 min	Min, Max, Avg & Median values
Response Time (in Milliseconds) – round trip time of E2E system	latency	Object	Every 5 min	Min, Max, Avg & Median values
Trading API Failure Count	failureTradeApi	Numeric	Every 5 min	Long value
Client Authentication Failure Count	failureAuthentication	Numeric	Every 5 min	Long value
Historical Requests/Second	historicalThroughput	Object	Start of Day	Min, Max, Avg & Median values
Historical Response Time (in Milliseconds)	historicalLatency	Object	Start of Day	Min, Max, Avg & Median values

Note: The metric key of Log monitoring has been deprecated, thus members do not have to send data for this parameter

-EACH METRIC SAMPLE PAYLOAD FOR EXAMPLE PURPOSE ONLY-

```
{
  "memberId": "09100",
  "exchangeId": 1,
  "sequenceId": 1,
  "locationId": 1,
  "timestamp": 1356998400000,

  "payload": [
    {
      "applicationId": -1,
      "metricData": [
        {
          "key": "throughput",
          "value": {
            "min": 10,
            "max": 50,
            "avg": 34,
            "med": 25
          }
        },
        {
          "key": "latency",
          "value": {
            "min": 10,
            "max": 50,
            "avg": 34,
            "med": 25
          }
        },
        {
          "key": "historicalThroughput",
          "value": {
            "min": 10,
            "max": 50,
            "avg": 34,
            "med": 25
          }
        },
        {
          "key": "historicalLatency",
          "value": {
            "min": 10,
            "max": 50,
            "avg": 34,
            "med": 25
          }
        }
      ]
    }
  ],
}
```

```

{
  "key": "failureTradeApi",
  "value": 2
},
{
  "key": "failureAuthentication",
  "value": 100
}
]
}
]
}

```

Response Object values

Sr. No.	Field Description	Field Name	Data Type	Description	Expected Values & Validations
1	Timestamp	timestamp	Numeric	Timestamp (in milliseconds since NSE epoch time i.e., 01-jan-1980)	1356998400000
2	Version No	versionNo	String	API Version	
3	Response Code	responseCode	Numeric	Application Processing status code	
4	Response Description	reponseDesc	String	Application Processing status description	
5	Errors	errors	List	List of error records in case of partial success Refer Error table	

In case of Success

```

{
  "timestamp": 1356998400000,
  "versionNo": "V1",
  "responseCode": 601,
  "responseDesc": "Success"
}

```

In case of Partial Success

```

{
  "timestamp": 1356998400000,
  "versionNo": "V1",
  "responseCode": 602,
  "responseDesc": "Partial Success",
  "errors": [
    {
      "applicationId": 1,
      "errCode": 707,
      "errDesc": "Invalid Metric Key"
      "errKey": "abc",
      "measure": null
    },
  ],
}

```

```

{
  "applicationId": 1,
  "errCode": 708,
  "errDesc": "Invalid Metric Value"
  "errKey": "latency",
  "measure": "min",
},
{
  "applicationId": 2,
  "errCode": 709,
  "errDesc": "Duplicate Record"
  "errKey": null,
  "measure": null
}
]
}

```

Hardware Metrics

This API will publish the different hardware metrics of the member end systems.

Method	API EndPoint	Header	Content-Type
POST	/metrics/hardware	Session Token	application/json

Request Object values

Field Description	Field Name	Data Type	Description	Expected Values & Validations	Optional/ Mandatory
Payload	payload	List	List of application records	List of payload	M
Exchange ID	exchangeId	Numeric	Different exchange ids	Refer Value table	M
Member ID	memberId	String	Member code	09100	M
Sequence ID	sequenceId	Numeric	Sequence number of the records	1	M
Location ID	locationId	Numeric (6)	Location Identifier number	1	O
Time Stamp (in milliseconds since NSE epoch time i.e., 01-jan-1980)	timestamp	Numeric	Time stamp of the records	1356998400000	M

Payload

Field Description	Field Name	Data Type	Description	Expected Values & Validations	Optional/ Mandatory
Application ID	applicationId	Numeric	Application Identification number	1	M

Metric Data	metricData	List	List of key value pair with all available metrics in key and their corresponding values	[{"key":"","value":""}] Available Keys: 1. cpu 2. memory 3. disk 4. uptime	M
-------------	------------	------	---	---	----------

Metric Data

Field Description	Key Name	Data Type	Frequency	Expected Value
CPU Utilization (in %)	cpu	Object	Every 5 min	Float (between 0-100, upto 2 decimals)
Memory Utilization (in %)	memory	Object	Every 5 min	Float (between 0-100, upto 2 decimals)
Disk Utilization (in %)	disk	Object	Every 5 min	Float (between 0-100, upto 2 decimals)
Uptime (in minutes)	uptime	Object	Every 5 min	Long value

```
{
  "memberId": "09100",
  "exchangeId": 1,
  "sequenceId": 1,
  "locationId": 1
  "timestamp": 1356998400000,
  "payload": [
    {
      "applicationId": 1,
      "metricData": [
        {
          "key": "cpu",
          "value": {
            "min": 0.10,
            "max": 50.10,
            "avg": 34.55,
            "med": 25.00
          }
        },
        {
          "key": "memory",
          "value": {
            "min": 0.10,
            "max": 50.10,
            "avg": 34.55,
            "med": 25.00
          }
        },
        {
          "key": "disk",
          "value": {
            "min": 0.10,
            "max": 50.10,
```

```

    "avg": 34.55,
    "med": 25.00
  }
},
{
  "key": "uptime",
  "value": {
    "min": 0,
    "max": 50,
    "avg": 34,
    "med": 25
  }
}
]
}
]
}

```

Response Object values

Sr. No.	Field Description	Field Name	Data Type	Description	Expected Values & Validations
1	Timestamp	timestamp	Numeric	Timestamp (in milliseconds since NSE epoch time i.e., 01-jan-1980)	135699840000
2	Version No	versionNo	String	API Version	
3	Response Code	responseCode	Numeric	Application Processing status code	
4	Response Description	reponseDesc	String	Application Processing status description	
5	Errors	errors	List	List of error records in case of partial success Refer Error table	

In case of Success

```

{
  "timestamp": 1356998400000,
  "versionNo": "V1",
  "responseCode": 601,
  "responseDesc": "Success"
}

```

In case of Partial Success

```

{
  "timestamp": 1356998400000,
  "versionNo": "V1",
  "responseCode": 602,
  "responseDesc": "Partial Success",
  "errors": [
    {
      "applicationId": 1,
      "errCode": 707,
      "errDesc": "Invalid Metric Key"
      "errKey": "abc",
    }
  ]
}

```

```

    "measure": null
  },
  {
    "applicationId": 1,
    "errCode": 708,
    "errDesc": "Invalid Metric Value"
    "errKey": "disk",
    "measure": "min",
  },
  {
    "applicationId": 2,
    "errCode": 709,
    "errDesc": "Duplicate Record"
    "errKey": null,
    "measure": null
  }
]
}

```

Database Metrics

This API will publish the different hardware metrics of the member end systems.

Method	API EndPoint	Header	Content-Type
POST	/metrics/database	Session Token	application/json

Request Object values

Field Description	Field Name	Data Type	Description	Expected Values & Validations	Optional/ Mandatory
Payload	payload	List	List of application records	List of payload	M
Exchange ID	exchangeId	Numeric	Different exchange ids	Refer Value table	M
Member ID	memberId	String	Member code	09100	M
Sequence ID	sequenceId	Numeric	Sequence number of the records	1	M
Location ID	locationId	Numeric (6)	Location Identifier number	1	O
Time Stamp (in milliseconds since NSE epoch time i.e., 01-jan-1980)	timestamp	Numeric	Time stamp of the records	1356998400000	M

Payload

Field Description	Field Name	Data Type	Description	Expected Values & Validations	Optional/ Mandatory
Application ID	applicationId	Numeric	Application Identification number	1	M

Metric Data	metricData	List	List of key value pair with all available metrics in key and their corresponding values	[{"key":"","value":""}] Available Keys: 1. status 2. qSize 3. throughput 4. latency	M
-------------	------------	------	---	--	----------

```
{
  "memberId": "09100",
  "exchangeId": 1,
  "sequenceId": 1,
  "locationId": 1
  "timestamp": 1356998400000,
  "payload": [
    {
      "applicationId": 1,
      "metricData": [
        {
          "key": "status",
          "value": 1
        },
        {
          "key": "qSize",
          "value": {
            "min": 0,
            "max": 50,
            "avg": 34,
            "med": 25
          }
        },
        {
          "key": "bandwidth",
          "value": {
            "min": 0.1,
            "max": 50.1,
            "avg": 34.55,
            "med": 25
          }
        },
        {
          "key": "latency",
          "value": {
            "min": 0,
            "max": 50,
            "avg": 34,
            "med": 25
          }
        }
      ]
    }
  ]
}
```

Metric Data

Field Description	Key Name	Frequency	Data Type	Expected Value
Replication Status (1=Up, 0= Down)	status	Every 5 min	Short	1
Replication Queue Size	qSize	Every 5 min	Object	Long value
Replication Bandwidth Utilization (in %)	bandwidth	Every 5 min	Object	Float (between 0-100, upto 2 decimals)
Replication Latency (in milliseconds)	latency	Every 5 min	Object	Long value

Response Object values

Sr. No.	Field Description	Field Name	Data Type	Description	Expected Values & Validations
1	Timestamp	timestamp	Numeric	Timestamp (in milliseconds since NSE epoch time i.e., 01-jan-1980)	1356998400000
2	Version No	versionNo	String	API Version	
3	Response Code	responseCode	Numeric	Application Processing status code	
4	Response Description	reponseDesc	String	Application Processing status description	
5	Errors	errors	List	List of error records in case of partial success Refer Error table	

In case of Success

```
{
  "timestamp": 1356998400000,
  "versionNo": "V1",
  "responseCode": 601,
  "responseDesc": "Success"
}
```

In case of Partial Success

```
{
  "timestamp": 1356998400000,
  "versionNo": "V1",
  "responseCode": 602,
  "responseDesc": "Partial Success",
  "errors": [
    {
      "applicationId": 1,
      "errCode": 707,
      "errDesc": "Invalid Metric Key",
      "errKey": "abc",
      "measure": null
    },
    {
      "applicationId": 1,
      "errCode": 708,
      "errDesc": "Invalid Metric Value",
      "errKey": "latency",

```

```

    "measure": "min",
  },
  {
    "applicationId": 2,
    "errCode": 709,
    "errDesc": "Duplicate Record"
    "errKey": null,
    "measure": null
  }
]
}

```

Network Metrics

This API will publish the different network metrics of the member end systems.

Method	API EndPoint	Header	Content-Type
POST	/metrics/network	Session Token	application/json

Request Object values

Field Description	Field Name	Data Type	Description	Expected Values & Validations	Optional/ Mandatory
Payload	payload	List	List of application records	List of payload	M
Exchange ID	exchangeId	Numeric	Different exchange ids	Refer Value table	M
Member ID	memberId	String	Member code	09100	M
Sequence ID	sequenceId	Numeric	Sequence number of the records	1	M
Location ID	locationId	Numeric (6)	Location Identifier number	1	O
Time Stamp (in milliseconds since NSE epoch time i.e. 01-jan-1980)	timestamp	Numeric	Time stamp of the records	1356998400000	M

Payload

Field Description	Field Name	Data Type	Description	Expected Values & Validations	Optional/ Mandatory
Application ID	applicationId	Numeric	Application Identification number	1	M
Metric Data	metricData	List	List of key value pair with all available metrics in key and their	[{"key": "", "value": ""}] Available Keys: 1. bandwidth 2. latency 3. packetCount 4. lookupCount	M

			corresponding values		
--	--	--	----------------------	--	--

Metric Data

Field Description	Key Name	Frequency	Data Type	Expected Value
Bandwidth Utilization (in %)	bandwidth	Every 5 min	Object	Float (between 0-100, upto 2 decimals)
Network Latency (in milliseconds)	latency	Every 5 min	Object	Long value
Packet Error Count	packetCount	Every 5 min	Numeric	Long value
DNS Lookup Failure Count	lookupCount	Every 5 min	Numeric	Long value

```
{
  "memberId": "09100",
  "exchangeId": 1,
  "sequenceId": 1,
  "locationId": 1,
  "timestamp": 1356998400000,
  "payload": [
    {
      "applicationId": 1,
      "metricData": [
        {
          "key": "bandwidth",
          "value": {
            "min": 0.10,
            "max": 50.10,
            "avg": 34.55,
            "med": 25.00
          }
        },
        {
          "key": "latency",
          "value": {
            "min": 0,
            "max": 50,
            "avg": 34,
            "med": 25
          }
        },
        {
          "key": "packetCount",
          "value": 25
        },
        {
          "key": "lookupCount",
          "value": 25
        }
      ]
    }
  ]
}
```

Response Object values

Sr. No.	Field Description	Field Name	Data Type	Description	Expected Values & Validations
1	Timestamp	timestamp	Numeric	Timestamp (in milliseconds since NSE epoch time i.e. 01-jan-1980)	1356998400000
2	Version No	versionNo	String	API Version	
3	Response Code	responseCode	Numeric	Application Processing status code	
4	Response Description	reponseDesc	String	Application Processing status description	
5	Errors	errors	List	List of error records in case of partial success Refer Error table	

In case of Success

```
{
  "timestamp" : 1356998400000,
  "versionNo" : "V1",
  "responseCode" : 601,
  "responseDesc" : "Success"
}
```

In case of Partial Success

```
{
  "timestamp": 1356998400000,
  "versionNo": "V1",
  "responseCode": 602,
  "responseDesc": "Partial Success",
  "errors": [
    {
      "applicationId": 1,
      "errCode": 707,
      "errDesc": "Invalid Metric Key"
      "errKey": "abc",
      "measure": null
    },
    {
      "applicationId": 1,
      "errCode": 708,
      "errDesc": "Invalid Metric Value"
      "errKey": "latency",
      "measure": "min",
    },
    {
      "applicationId": 2,
      "errCode": 709,
      "errDesc": "Duplicate Record"
      "errKey": null,
      "measure": null
    }
  ]
}
```

```

    ]
  }

```

Response Codes

Status Codes

Codes	Description
200	Successful response body.
400	Bad request
401	Unauthorised Token
403	Forbidden if the user does not have permission.
405	Method not found
500	Internal Server Error
503	Service Unavailable

Validation Error Codes and Description

Validation Error Codes	Validation Messages	Description
601	Success	When the API is a Success
602	Partial Success	When partial records in an API are successful
603	Invalid Payload	Entire data entered in the payload is invalid
605	Something went wrong	It's a technical error
607	Concurrent User limit exceeded	Multiple API requests triggered together, and the API call limit exceeded
701	Invalid member id or login id or password	For Log In API
702	User is disabled	User is disabled
703	Invalid member id or login id	For Log Out API
704	Invalid Sequence ID, the sequence ID should be "X". Where, X is the actual sequence id that is to be entered by member.	The sequence id entered is invalid, incorrect or is duplicate.
705	Invalid timestamp	If the data to be pushed by members is before an interval of 5 mins from the last time when the data was pushed
706	Invalid request	Request made for any API is invalid
707	Invalid Metric Key	Invalid metric key in Payload
708	Invalid Metric Value	Invalid metric value in Payload
709	Duplicate record	Duplicate records for an Application ID
710	Payload size exceeded	Payload has more than 5 records
711	Invalid Member ID	Member ID entered is invalid
712	Invalid Exchange ID	Exchange ID entered is invalid
713	Invalid Application ID	Application ID entered is invalid
714	Invalid Location ID	Location ID entered is invalid (negative and null values) or user has passed the field of LocationId, but no value is passed against this field i.e. it is blank.
801	Invalid token	For all APIs
802	Authentication token expired	For all APIs

901	Not valid record	Blank or null values are passed for more than one field of the request body (for all APIs except Login and Logout)
902	Invalid version no	For all APIs
904	Invalid URL	URL name is incorrect
905	Invalid content type	Invalid request body data type
906	Authentication server down	Authentication server down
907	Password attempt Limit Exceeded	Multiple attempts made with incorrect password

Annexure

Attached is an excel defining each metric key along with its frequency i.e. time intervals at which its value is to be published via the APIs.



LAMA - Metric Key
Details.xlsx

Version History

Version History		
Version	Release date	Change Description
1.0	10/02/2023	Base version
1.1	06/03/2023	- Point no 3 added in the Notes under Section LAMA APIs - Section Login and Logout – In the table ‘Response JSON values’ parameters ‘Timestamp’ and ‘Version No’ added - Section Login and Logout – Under the header ‘Response Object Structure (JSON)’ example updated for parameters ‘Timestamp’ and ‘Version No’. - Section Login and Logout – In the table ‘Response JSON values’ parameter ‘Status’ replaced by parameter ‘Response Description’. - Section Login and Logout – Under the header ‘Response Object Structure (JSON)’ example updated for parameter responseDesc. - Section Error Table – Field ‘Error Description’ added - New section as Log Table added to define the log parameters - Section Application Metrics – Under header ‘Metric Data’ description for the field Log updated. - Section Application Metrics under the head In case of Partial Success – example updated for new field ‘Error Description’ (errDesc). - Section Hardware Metrics under the head In case of Partial Success – example updated for new field ‘Error Description’ (errDesc). - Section Database Metrics under the head In case of Partial Success – example updated for new field ‘Error Description’ (errDesc). - Section Network Metrics under the head In case of Partial Success – example updated for new field ‘Error Description’ (errDesc). - Section Status Code – Status Code 503 added - Section Validation Error Codes and Description – Error codes 701 and 703 updated for validation messages wherein Member Code replaced by Member ID. - Section Validation Error Codes and Description – Error code 704 updated for validation message and description. - Section Validation Error Codes and Description – Error code 710 (Payload size exceeded) description updated as ‘Payload has more than 5 records’ - Section Validation Error Codes and Description – Error code 901 description updated as ‘Blank or null values are passed for more than one field of the request body (for all APIs)’ instead of ‘Blank or null values not allowed’ - Section Validation Error Codes and Description – Error codes 606 (Method Unknown), 903 (Invalid Syntax) and 908 (API call not allowed) removed. - Section Validation Error Codes and Description – Error codes 603, 711, 712, and 713 added - New section as Annexure added defining the metric keys
1.2	24/03/2023	- Section LAMA APIs - Base URL for LAMA API updated. - Section LAMA APIs – UAT URL for all the API endpoints added - Section Login – Under header ‘For Password’ a document containing code snippets for password encryption has been added.
1.3	04/10/2024	- New parameter as Location Id has been introduced across Application, Database, Network and Hardware metrics. - Error code as 714 for Invalid Location ID introduced in the Validation

		• Error Codes and Description section • Notes added for metric keys deprecated like Log Monitoring.
1.3	18/02/2026	• Latency value updated to milliseconds from microseconds